# Step 8 summary - final mock CSC662 paper vs 3 clean modern papers

**Scope.** mock_paper_04.pdf (6 pages), built from the final hardened realistic run seed 12 (DeepSeek V4.1 Flash), vs JUN 2019 (7), DEC 2019 (6), FEB 2022 (14). Word COM export; 200 dpi renders. Full report: step8_comparison.md. Paper-01 version archived as step8_comparison_paper01.md.

## Structure

| Feature | Mock 04 | JUN 2019 | DEC 2019 | FEB 2022 |
|---|---|---|---|---|
| Part A / B | 10 x (2+6) = 80 / 2 x (5+5) = 20 | same | same | same |
| Total / duration | 100 / 3 HOURS | 100 / 3 HOURS | 100 / 3 HOURS | 100 / 3 HOURS |
| Part headings | PART A (80 MARKS) / PART B (20 MARKS) | same | same | same |
| Cover | FINAL EXAMINATION, 5 instructions | same style | same style | FINAL TEST, 3 instructions + name/marks grid |
| Printed pages | 6 | 7 | 6 | 14 |
| Computed / theory marks | 6 / 94, 0 placeholders | 6 / 94 | 6 / 94 | 0 / 100 |

**Matches:** parts, question counts, 2/6/5/5 marks, 100 total, 3 hours, a/b numbering, part mark totals, header/footer on every body page, right-aligned mark labels. **Differences:** 6 vs 7/14 pages (matches DEC); cover differs from FEB 2022; computed share differs from FEB.

## Layout (pages 2+)

| | Mock 04 | JUN 2019 | DEC 2019 | FEB 2022 |
|---|---|---|---|---|
| Header / footer present | 5/5 / 5/5 | 6/6 / 6/6 | 5/5 / 5/5 | 13/13 / 13/13 |
| Header / footer / body ink (median) | 0.0240 / 0.0070 / 0.0428 | 0.0038 / 0.0102 / 0.0368 | 0.0116 / 0.0105 / 0.0459 | 0.0125 / 0.0166 / 0.0184 |
| Question heading | 10.7 pt bold, x=0.123 | OCR only | 9.7-10.0 regular, x=0.136 | 11.0 bold, x=0.121 |
| a/b labels / marks labels | 24 / 24 | 24 / 24 (OCR) | 24 / 24 | 25 / 24 |
| Mark right edge / page width | 0.879 | OCR only | 0.868 | 0.879 |

Body ink is now close to DEC (0.0428 vs 0.0459), so the 6-page paper matches DEC density. Header ink is heavier than all three real papers (bold header); footer ink is lighter. FEB 2022 gives each question a page; the mock is tighter and has less answer space.

## Computed share

Mock: **6 computed marks (1 Vigenere slot) = 6 %**, 94 theory marks across 23 sourced questions. Clean modern fingerprint: **40 machine-verifiable marks over 10 papers = 4.0 marks/paper**; the three comparison papers carry 6, 6 and 0. The paper is realistic, not practice mode, and supersedes paper 01's 28 computed marks / 28 %. Only one cipher type, so no repeated Caesar/Playfair slots.

## Theory content and wording

23/23 theory questions accepted; 94 marking points; 0 placeholders; source overlap 0; max near-copy **0.614** (threshold 0.80). Low stem-overlap: **36/94 (38.3%)**; entailment initial yes 22 / partly 12 / no 2, both no points regenerated; final yes 23 / partly 13 / no 0. Verbatim-copy rejections: 6. Both Part B groups have a shared scenario (grounding 0.391), versus the measured 7/19 = 36.8 % rate.

| Marks | Mock median words (range) | Clean modern median (p90) |
|---|---|---|
| 2 | 20 (18-30) | 9 (18) |
| 5 | 31.5 (24-36) | 26 (43) |
| 6 | 31 (28-45) | 18 (44) |

Generated 2-mark stems are the main wording gap (20 vs 9 words); 6-mark stems are also longer (31 vs 18). Command mix this seed: Explain 11, State 6, Describe 2, Discuss 2, List 2; no Compare/Propose/Analyse/Justify. Closest past pair: A1a ethics vs CSC662-2018-06 Q15a = 0.614. The stems are grounded and non-copied, but wordier and more scaffolded than real questions.

## Marking scheme

mock_scheme_04.pdf, 16 pages, same header/footer as the paper: computed Vigenere working + 26 x 26 tableau, every theory point with verbatim quote and slide citation, *supported by lecture slides (not lecturer-marked)* label, stem-overlap scores and low-overlap flags. No official CSC662 schemes exist; this is far more detailed than a real scheme.

## Weakness list (specific)

- 2-mark stems too long: 20 vs 9 words median; 6-mark 31 vs 18.
- Command mix narrow in this seed: no Compare/Propose/Analyse/Justify.
- Header ink 0.024 vs 0.012 real (bold running header); footer lighter (0.007 vs 0.010-0.017).
- Body density close to DEC but much denser than FEB; 6 vs 14 pages, less answer space.
- 36/94 marking points (38.3%) remain low stem-overlap; accepted by entailment but flagged for review.
- Both Part B groups have scenarios (over-represents the 36.8 % measured rate).
- No official marking scheme; 16-page scheme is far more detailed than real.
- Theory answers are slide-supported, not lecturer-marked.
- Computed share 6 % vs 4.0-mark modern average (within range but 1.5x).
- Cover matches JUN/DEC but differs from FEB 2022.